# LiDAR Point Cloud Denoising for Autonomous Driving Applications

## Technical Summary of Implementation Results

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## 1. System Implemented

The pipeline proposed in Section 5 of the project proposal has been implemented in full. Raw KITTI Velodyne scans are preprocessed (NaN/Inf removal, 60 m range gating, 0.15 m voxel-grid downsampling) and corrupted by a physics-based weather model: extinction follows an empirical Mie fit $\alpha = 0.003R^{0.6}$ m$^{-1}$ for rain rate $R$ (mm/h) at 905 nm, after LISA [Kilic et al., arXiv:2107.07004]. Each return is attenuated by the two-way transmittance $e^{-2\alpha r}$; a droplet intercepting the beam relocates the return to its own (nearer) range, weakly attenuated echoes fall below the detection threshold and are *deleted*, and survivors receive SNR-dependent range jitter. Gaussian sensor jitter ($\sigma = 0.05$ m) and uniform outliers (8%) are applied on top, and three severity presets (10/25/50 mm/h) follow the Robo3D convention [Kong et al., ICCV 2023].

The deletion channel matters for how every result below should be read: a denoiser can remove spurious points and reposition jittered ones, but it cannot restore a return the weather destroyed. Reconstruction error therefore has a strictly positive floor, quantified in Section 2.

Six denoisers are evaluated. Three require no training: (a) the statistical baseline chaining Statistical Outlier Removal ($k = 20$, $\alpha = 2.0$) and Radius Outlier Removal (4 neighbours within 0.6 m); (b) IDSOR [arXiv:2602.05876], which normalises the $k$-NN distance statistic by range so far-field structure is not eroded and then tightens the threshold for weak returns; and (c) a temporal-consistency filter keeping only points with support in the previous sweep, preceded by ICP ego-motion compensation. Three are learned: (d) a PointNet autoencoder (0.83M parameters, Chamfer loss), (e) a lightweight DGCNN (0.22M parameters) regressing per-point displacements and classifying injected noise, and (f) *RangeRestoreNet*, a dual-head U-Net over a $64 \times 2048$ range image (1.94M parameters)

developed here, which classifies noise per pixel and predicts a bounded per-pixel range correction.

Intensity is carried throughout. Weather backscatter returns markedly less energy than a solid surface at the same range — measured here, real returns have median intensity 0.150 against 0.051 for injected noise — and methods (b) and (f) exploit that; the earlier geometry-only pipeline discarded the channel entirely.

**Data.** 262 scans from two public KITTI raw drives (2011_09_26, drives 0001 and 0005), split 213/26/28 into train/validation/test by frame. The official KITTI 3D-object Velodyne archive (29 GB, login-gated) was deliberately not used: the raw split provides identical sensor data at 490 MB, sufficient for denoising experiments. Training used 28,510 patches of 256 points with 3,415 held-out validation patches; on team hardware (Apple M1, MPS backend) both models were trained for 10 epochs, and the checkpoint retained is the one minimising *validation* loss rather than the final epoch.

# 2. Reconstruction Quality (Proposal Table-1 axes)

Mean ± standard deviation over 10 held-out frames × 3 independent noise draws (30 paired observations, moderate severity, 25 mm/h). CD = Chamfer Distance, $HD_{95}$ = 95th-percentile Hausdorff. Every pairwise CD difference below is significant at $p < 10^{-5}$ (Wilcoxon signed-rank over the paired observations).

| **Method** | **CD (m)** | **$HD_{95}$** | **Oracle gap** | **Rem. F1** | **Reten.** |
|---|---|---|---|---|---|
| Noisy input | 0.572 ± 0.111 | — | 100% | — | — |
| *Oracle (perf. removal)* | *0.104 ± 0.003* | *0.353* | *0%* | *1.000* | *68.9%* |
| DGCNN | **0.111 ± 0.008** | **0.342** | **1.6%** | 0.788 | 78.7% |
| IDSOR | 0.174 ± 0.013 | 0.827 | 15.0% | 0.665 | 81.3% |
| RangeRestoreNet | 0.215 ± 0.035 | 0.459 | 23.8% | **0.814** | 65.6% |
| SOR + ROR | 0.251 ± 0.020 | 1.529 | 31.5% | 0.628 | 78.7% |
| Temporal + SOR | 0.317 ± 0.035 | 2.173 | 45.5% | 0.567 | 81.4% |
| PointNet AE | 0.521 ± 0.092 | 3.353 | 89.1% | 0.000 | 100.0% |

The *oracle gap* column is the fraction of the *recoverable* error a method leaves behind, $(\mathrm{CD}_{\mathrm{method}} - \mathrm{CD}_{\mathrm{oracle}})/(\mathrm{CD}_{\mathrm{noisy}} - \mathrm{CD}_{\mathrm{oracle}})$. It is reported because a raw Chamfer Distance is uninterpretable when 31% of returns are destroyed beyond recovery: 0% means a method matched a perfect noise classifier, 100% that it achieved nothing.

The oracle row is the reference point: it deletes every injected point and keeps every real one, and still scores CD 0.104 rather than zero, because the weather model destroys 31% of returns that no denoiser can restore. Read against that floor, the DGCNN has closed 98% of the recoverable gap.

The DGCNN is the strongest method by a wide margin, and marginally exceeds the oracle on $HD_{95}$ (0.342 vs 0.353) — a difference of 3% on a single tail statistic, which should not be over-read. An earlier revision of this work attributed it to the DGCNN's displacement head correcting jitter the oracle leaves untouched; direct measurement (Section 3) shows that head changes point positions by 2.6% and alters Chamfer Distance not at all, so that explanation does not hold.

RangeRestoreNet achieves the best noise-removal F1 of any method (0.814, above the DGCNN's 0.788) at 40× lower latency, but a worse Chamfer Distance, because it is recall-heavy: it finds more of the noise and removes real returns along with it, retaining only 65.6% of the cloud — below even the oracle's 68.9%.

The statistical baseline remains competitive and far cheaper (Section 4), consistent with the literature survey's expectation that classical filters handle additive outliers well; its weakness is recall (removal F1 0.628, recall 0.53), as a global distance threshold cannot separate weather backscatter from sparse genuine structure at range. The temporal filter under-performs the plain baseline here: requiring support in the previous sweep also discards legitimately new returns at the leading edge of the scene.

The autoencoder under-performs on every axis: a global bottleneck over-smooths fine scan-line structure, and having no removal head it cannot delete an injected point at all (removal F1 exactly 0). Its retention of 100% confirms it is a pure displacement model. Section 3 shows this is far more damaging than its modest CD suggests.

**Ablations.** Sweeping the DGCNN's outlier threshold over 0.3–0.7 confirms the default of 0.5 sits at the CD optimum (0.113, flat between 0.5 and 0.6) with removal F1 peaking slightly earlier at 0.4 (0.788); the model is not sensitive to this choice. Sweeping SOR parameters confirms the proposal's tuning discussion: tighter thresholds ($\alpha = 1.0$) improve CD by ∼6% over the default. ICP ego-motion compensation improves the temporal filter from CD 0.227 to 0.177 while retaining 5 percentage points more of the cloud, confirming that the uncompensated variant was deleting static structure rather than backscatter.

# 3. Downstream Detection Impact

KITTI raw frames carry no ground-truth boxes, so a *detection consistency* protocol was used: a KITTI-pretrained PointPillars detector (MMDetection3D, CPU inference) is run on each clean cloud and its boxes taken as reference; recall/precision of the detections on noisy and denoised clouds are measured against that reference (BEV centre matching, 1 m tolerance, score $\geq 0.4$, 10 frames).

Evaluated over all 28 held-out frames × 3 noise seeds (84 paired observations). $p$-values

are paired Wilcoxon signed-rank against the perfect-removal oracle.

| **Condition** | **Recall** | **Precision** | $p$ **vs oracle** |
|---|---|---|---|
| Noisy input | $0.259 \pm 0.231$ | 0.806 | $< 10^{-4}$ |
| *Oracle (perfect removal)* | *0.479 ± 0.325* | *0.805* | — |
| DGCNN | $\mathbf{0.452 \pm 0.298}$ | 0.764 | 0.153 (n.s.) |
| RangeRestoreNet | $0.403 \pm 0.320$ | 0.788 | $2 \times 10^{-4}$ |
| SOR + ROR | $0.392 \pm 0.303$ | 0.737 | $< 10^{-4}$ |
| Temporal + SOR | $0.389 \pm 0.269$ | 0.795 | — |
| IDSOR | $0.368 \pm 0.291$ | **0.826** | $< 10^{-4}$ |
| PointNet AE | $0.062 \pm 0.104$ | 0.773 | $< 10^{-4}$ |

The weather corruption costs the detector 74% of its clean-cloud detections. Every denoiser recovers a significant share ($p < 10^{-4}$ against the noisy baseline) except the autoencoder, which makes matters significantly worse.

**Perfect removal is the ceiling.** The DGCNN is the only method statistically indistinguishable from the oracle; all others fall significantly short. This sample size matters: an earlier revision of this work evaluated detection on 10 frames, observed that two methods exceeded the oracle's mean recall, and drew the opposite conclusion. Per-frame recall has a standard deviation of $\approx 0.3$ about a mean of 0.4, so at that sample size a gap of 0.05 between means is uninformative; at 84 paired observations the ordering reverses and the apparent effect disappears. The geometric metrics, whose between-frame variance is an order of magnitude smaller, did not suffer this problem — which is precisely why the same sample size was wrongly assumed adequate for both.

One further result deserves emphasis.

**Position restoration does not contribute.** Two direct measurements. The DGCNN's displacement head changes the mean position error of surviving real points from 0.0804 m to only 0.0783 m, and removing the displacement entirely leaves Chamfer Distance unchanged (0.1158 vs 0.1157); a range-restoration head trained for the same purpose on the range-image model (Section 4) converged to predicting exactly zero. The reason is intrinsic rather than a training failure: weather-induced range jitter is zero-mean, so for a single observation the error-minimising prediction *is* zero, and recovering it requires a surface prior over neighbouring returns that neither a per-point patch loss nor a per-pixel image loss supplies. Position restoration on this corruption is, on the present evidence, not a source of downstream gain.

**Reconstruction error does not predict perception quality.** The autoencoder improves Chamfer Distance by 8.9% relative to the noisy input yet destroys the detector, dropping recall from 0.259 to 0.062 — a 76% loss, $p < 10^{-4}$, and the only condition that is worse than no denoising at all. A second, milder inversion points the same way: IDSOR

achieves clearly better Chamfer Distance than SOR+ROR (0.174 vs 0.251) yet slightly lower detection recall (0.368 vs 0.392, $p = 0.069$). Across all six denoisers the rank correlation between Chamfer Distance and detection recall is only $\rho = -0.60$ ($p = 0.21$) — directionally correct, statistically weak. The effect cannot be attributed to point-count inflation, because after the recombination correction of Section 6 the AE emits exactly 100% of the input cardinality; it is the oversmoothing itself that removes the structure PointPillars relies on. The recombination ablation supplies a second, independent demonstration: the uncorrected variant achieves a *better* CD (0.525 vs 0.569) while returning 232% of the input points, because densifying a cloud shortens the clean→denoised nearest-neighbour term faster than it lengthens the reverse term. Chamfer Distance is thus improvable both by blurring and by duplication, neither of which helps perception. This directly substantiates the research gap identified in Section 3.5 of the proposal, and motivates reporting retention and a downstream task metric alongside any reconstruction figure.

# 4. Real-Time Feasibility (10 Hz target)

Per-frame latency was profiled end-to-end on the Apple M1 (batch inference, patch covering, k-d tree construction included). Dynamic int8 quantisation was applied to the learning models as the Step-5 optimisation.

| **Method** | **ms/frame** | **Hz** | **≥10 Hz** |
|---|---|---|---|
| RangeRestoreNet (MPS) | **45** | **22.1** | **yes** |
| IDSOR (CPU) | **49** | **20.3** | **yes** |
| SOR+ROR (CPU) | **65** | **15.5** | **yes** |
| Temporal+SOR (CPU, incl. ICP) | 278 | 3.6 | no |
| PointNet AE (MPS) | 1,192 | 0.84 | no |
| DGCNN (MPS) | 1,806 | 0.55 | no |
| DGCNN (CPU int8) | 11,593 | 0.09 | no |

**The 10 Hz target is met, and for the first time by a learned model.** RangeRestoreNet runs end-to-end — projection, inference and unprojection — in 45 ms, a 40× speedup over the DGCNN. The gain comes entirely from the representation: a rotating LiDAR is natively a 2-D array of (laser channel, azimuth) samples, so projecting to a range image turns neighbourhood queries into array indexing and lets 2-D convolutions supply the context that EdgeConv had to compute by building a $k$-NN graph per patch. This is the change the recent literature converged on, and it is worth roughly two orders of magnitude.

Two caveats keep the classical numbers honest. Part of SOR+ROR's headroom is supplied by the corruption itself: precipitation destroys 31% of returns, so the cloud reaching the filter is smaller than a clear-weather scan, and the same pipeline is slower in good conditions — when it is least needed. And all timings are measured end-to-end from a NumPy array, including the Open3D conversion that earlier revisions excluded from the baseline's timer.

The accuracy/latency trade-off is therefore genuine but no longer stark: the DGCNN buys the lowest Chamfer error (oracle gap 1.6%) at 40× the cost of RangeRestoreNet, which in turn achieves the best noise-removal F1 of any method and recovers more detection recall than either statistical filter. Dynamic int8 quantisation reduced the AE checkpoint 3.4× (3.34→0.98 MB), valuable for the proposal's embedded Raspberry Pi scenario, but did not improve latency: PyTorch's dynamic backend accelerates Linear layers, whereas the bottleneck is the $k$-NN gather and patch assembly.

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# 5. Supervision Without Ground Truth

Every learned denoiser above is trained on the noise labels emitted by the corruption model itself — information no deployed system possesses, and which cannot be obtained from real data, since the same scene cannot be recorded simultaneously with and without rain. Whether the approach survives without those labels is therefore the decisive question for its practical relevance, and it is tested directly here.

Following LIORNet [arXiv:2603.19936], per-point pseudo-labels are derived from the physics of a weather return using only a single corrupted scan: (i) a *range-conditioned intensity anomaly* — received power falls as $\rho/r^2$, so the median intensity is estimated per range bin from the scan itself and a return is anomalous only relative to its own range, a global threshold would merely relabel "far" as "noise"; (ii) *neighbourhood range coherence* — a real surface is continuous in the range image whereas a droplet floats in front of what it occluded; and (iii) *edge protection* — genuine silhouettes also produce discontinuities, but coherent ones persisting across adjacent pixels. LIORNet's fourth

rule uses zero-reflectivity, which Ouster sensors report and the KITTI HDL-64E does not, so it is omitted rather than approximated. The restoration head is disabled in this configuration: its target is the clean range of a known source point, which is ground truth by construction.

| **Variant** | **CD (m)** | **$HD_{95}$** | **Removal F1** | **Reten.** |
|---|---|---|---|---|
| Pseudo-label rules alone (no network) | 0.465 | 3.297 | 0.484 | 70.5% |
| Self-supervised network | 0.436 | 2.564 | 0.491 | 67.7% |
| Supervised network | **0.227** | **0.460** | **0.813** | 65.3% |

The pseudo-labels themselves score precision 0.500 / recall 0.470 / F1 0.484 against the true labels: roughly half of what the self-supervised model is told is wrong. Trained on them, the network reaches 60% of the supervised model's removal F1 — but essentially all of that is supplied by the rules rather than by learning, since the network moves F1 only from 0.484 to 0.491 (+1.4% relative). It reproduced its supervision instead of generalising past it, and reporting the ratio to the supervised model alone would credit the network for what three physical heuristics already deliver.

Learning does help in one respect: $HD_{95}$ improves from 3.297 to 2.564 (22%) and CD from 0.465 to 0.436, so the network is better than the rules at catching the most damaging surviving outliers even though its average classification accuracy is not.

**This is a negative result for self-supervision as implemented, and a useful one:** the gap between F1 0.49 and 0.81 measures exactly what knowledge of the ground truth is worth on this task, and no additional training on these rules closes it. The prerequisite for progress is better pseudo-labels — multi-echo returns, temporal consistency across successive sweeps, or the reflectivity channel this sensor does not provide — not longer training.

# 6. Experimental Validity

**Protocol.** Every method is run on the *identical* noisy cloud for a given (frame, seed) pair, over three independent noise draws across ten held-out frames. Results are therefore paired, and each ranking claim is supported by a Wilcoxon signed-rank test over the 30 paired observations (recorded in `results/metrics_summary.json`); dispersion is reported as a standard deviation alongside every mean. Reconstruction quality is reported as Chamfer Distance and $HD_{95}$; strict Hausdorff is retained only in the raw output, because as the maximum over all residual points it is pinned near 30 m for every method by a single survivor at the range gate and consequently ranks nothing. Because geometric error alone cannot distinguish removing noise from displacing everything slightly,

methods that yield an index set into the noisy cloud (both filters, and the DGCNN via its outlier head) additionally report noise-removal precision, recall and F1 against the ground-truth injection labels.

**Corrections applied during review.** Six defects affecting reported numbers were identified and fixed; they are recorded here because several bear directly on the conclusions.

1. *Autoencoder patch recombination.* The AE's unordered reconstructions were concatenated across overlapping patches and voxel-downsampled, returning 165% of the input point count — inflating its Chamfer term from the denoised side and presenting the detector with a cloud of the wrong density. Each input point now adopts its nearest reconstructed point within its own patch, averaged across overlaps, giving output cardinality equal to input and a like-for-like comparison with the DGCNN.
2. *Autoencoder training target.* The Chamfer target was the nearest-clean-point projection of *every* noisy point, outliers included, so roughly 15% of the target set consisted of surface points duplicated wherever outliers happened to fall. The target is now a genuine clean patch gathered about the patch centroid.
3. *Temporal filter ego-motion.* Inter-sweep translation was assumed negligible against the 0.5 m consistency radius. Measured by ICP over these drives it averages 0.56 m and reaches 0.94 m (3–9 m/s), exceeding the radius on half of all frames; uncompensated, the filter deletes static structure rather than backscatter. Registration now precedes the consistency test.
4. *Patch coverage.* Randomly seeded patch centres left 2,147 of 36,033 points (6%) outside every patch on a representative frame. Such points retained an outlier probability of zero and could never be removed, capping the DGCNN's removal recall independently of classifier quality. Coverage is now completed greedily, at a cost of 27% more patches (+38 ms/frame) — a latency regression accepted deliberately.
5. *Filter timing.* SOR and ROR started their timers after the NumPy-to-Open3D conversion while the learned methods were timed end-to-end, flattering the baseline in the latency comparison. All methods are now timed end-to-end from a NumPy array.
6. *Degenerate frames.* An aggressive filter can empty a sparse frame; the metrics returned infinity, which propagated silently into averaged tables. Such frames now yield NaN, are excluded from means, and are counted explicitly in the results table.

Corrections (1) and (2) bear on the project's central claim that Chamfer Distance is

a poor proxy for downstream detection quality, since that claim rested largely on the autoencoder's collapse and the autoencoder was handicapped by two defects unrelated to its architecture. Section 3 reports the claim as re-measured after the fixes.

**Remaining limitations.** The noise is synthetic: the LISA model is a physically motivated approximation, not measured adverse-weather data, and validation on a real corrupted dataset (e.g. STF or the Canadian Adverse Driving Conditions set) remains the natural next step. The detection-consistency protocol measures agreement with the detector's own clean-cloud output rather than ground truth, so it bounds recoverable performance without certifying absolute accuracy. The intensity channel is zero-filled before detection. Finally, both learned models were trained for 10 epochs on laptop hardware; the DGCNN's validation loss was still decreasing slowly at that point, so its results should be read as a floor on what the architecture achieves rather than a converged ceiling.

# 7. Deliverables and Scope Notes

The repository contains the modular pipeline (`src/lidar_denoise/`), a YAML-configured driver (`scripts/run_pipeline.py`), the detection evaluation (`scripts/detection_eval.py`), an interactive before/after demo (`scripts/demo.py`), and 23 unit tests covering preprocessing, the weather-model physics, the filters, ego-motion registration, the metrics and degenerate-input handling. All metrics and plots are written under `results/`, alongside the corrected proposal source.

The YDLIDAR X2 and Raspberry Pi live-capture component of the proposal is hardware-dependent and remains deferred; the pipeline is sensor-agnostic and accepts any $(N, 3)$ scan, so a driver for that sensor can replace the KITTI reader without further changes.